JOSE DAVID RODRIGUEZ PINILLA

Full Stack Developer & IoT | Systems Engineering and Telematics Engineering Student · Universidad ICESI

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Available in Cali: on-site · hybrid · remote

PROFESSIONAL SUMMARY

Dual-degree student (Systems Engineering and Telematics Engineering) at Universidad ICESI and full stack developer. Co-founder of Fuzyum Studio, a custom software studio, and freelance developer of websites and management software for businesses in Cali. I build complete solutions: web applications with TypeScript, React, Next.js and NestJS; backends with Python (FastAPI), Java (Spring Boot) and Laravel on PostgreSQL; and IoT systems that integrate ESP32 hardware, real-time MQTT telemetry and Docker containers. I have shipped projects to production with Docker, Nginx, Coolify and GitHub Actions. In telecommunications I work with Cisco Packet Tracer and MATLAB. Active member of IEEE ICESI, ComSoc and the Astrobiology Research Group. Looking for full stack, backend or connected-systems opportunities where I can contribute technical value, with the ability to solve problems autonomously. English level: B1.

TECHNICAL SKILLS

Languages: TypeScript · JavaScript · Java · Python · SQL · PHP · C++ (ESP32/PlatformIO) · R

Backend & Databases: NestJS · Next.js · FastAPI · Spring Boot · Laravel · PostgreSQL · Drizzle · TypeORM · SQLAlchemy · InfluxDB

Frontend & Web: React · Next.js · Astro · Vite · Tailwind · HTML · CSS · GSAP · PWA · Wix

Telecommunications & Networking: Cisco Packet Tracer (IPv4/IPv6 subnetting, routing protocols, VLANs) · MATLAB · MQTT

IoT, Data & Automation: ESP32 · Node-RED · scikit-learn · Pandas · Gemini API · Automation Anywhere (bootcamp)

DevOps & Tools: Docker · Docker Compose · Nginx · Coolify · GitHub Actions · Git · Playwright · Vitest

WORK EXPERIENCE

Co-founder & Full Stack Developer *Oct 2026 – Present*

Fuzyum Studio (in incorporation) · Cali, Valle del Cauca · Hybrid · fuzyum.com

- Co-founded, with a partner, a studio for custom software, websites and automation. Besides development, I handle the administrative and sales side.
- Built most of fuzyum.com with Next.js 16, TypeScript and Tailwind: contact form, reference pricing, animated hero and a Docker image with Nginx deployed on Coolify, with cookie-free analytics.
- On ZymPOS, an offline-capable point of sale (Laravel 13 and PostgreSQL API, Tauri 2, Rust and Svelte desktop register), I closed the V1 release: business time zone handling, sales sync that keeps the register time, plans and pricing, a Windows installer and deployment from CI.
- ZymPOS V1 shipped with CI green: 1,180 Laravel tests, 495 Rust tests and 179 Vitest tests.

Freelance Software Developer *Oct 2026 – Present*

Self-employed · Cali, Valle del Cauca · On-site and remote · josed235.github.io

- Design and build websites for local businesses in Cali (retail, restaurants and gyms) with Astro, Vite and GSAP: catalogs with WhatsApp ordering, "open now" business hours, local SEO and Google Maps listings. Published case: the Calipieles landing page.
- Building a field service order system for technical maintenance companies (Next.js 16, React 19, PostgreSQL 17, Drizzle and Better Auth): technician schedule on mobile, equipment service history, checklists, photos, customer signature, PDF reports and a multi-tenant architecture.
- Developing a SaaS for gyms with an admin panel and a member app (PWA with QR check-in, routines and plans), with end-to-end tests in Playwright.

Front-End & Web Development Intern *Jun 2026 – Aug 2026*

Fundación Algo en Común · Part-time · Cali, Valle del Cauca · Hybrid

- Led the redesign, development and updating of the institutional web platform, optimizing the user interface and navigation experience to boost the organization's digital impact.
- Managed page configuration, responsive layout design and visual component optimization to ensure optimal performance across multiple devices.
- Collaborated on digital content administration and technical support for the websites backing the foundation's communication initiatives (Wix).

Sales Assistant *Dec 2022 – Feb 2023 and Dec 2023 – Feb 2024*

Calipieles · seasonal contract

- Customer service and cash register handling on seasonal contracts (2 seasons).

Laboratory Assistant *Aug 2022 – Jan 2023*

Fablab Cali, Universidad Autónoma de Occidente

- Assisted students and faculty; managed and maintained equipment in the digital fabrication lab.

Administrative Assistant *Mar – May 2022*

Center for Academic Excellence, Universidad Autónoma de Occidente

- Administrative support and assistance to students and faculty.

FEATURED PROJECTS

Camino Legal – Legal assistant for rural Colombia *Oct 2026*

LINXCO Hackathon · Team of 2 · caminolegal.space

- Built most of a progressive web app that helps rural communities draft formal petitions (derechos de petición) and review contracts with a risk traffic light, in plain language and without needing a signal.
- Modular monolith: NestJS API and React + TypeScript PWA sharing the conversation state machine, with WhatsApp and SMS channels on the same engine and an AI drafting assistant (Gemini).
- Offline-first: stores cases in IndexedDB and syncs them idempotently when the connection returns.
- Security with AES-256-GCM encryption, HMAC, role-based JWT and PIN lockout. Tests with Vitest, Playwright and axe on GitHub Actions. Deployed it myself at caminolegal.space with Docker and Nginx.

Capstone Project 1 · IoT Tremor Measurement System (Archimedes Project)

Individual project · Universidad ICESI · Repository: ProyectoIntegrador1-Tel

- Designed and implemented a complete IoT ecosystem to capture motion and pressure data during the clinical Archimedes spiral test, integrating MPU6050, LSM303DLHC and FSR402 sensors on an ESP32 (I2C / 12-bit ADC) sampling at 100 Hz.
- Built an asynchronous telemetry pipeline over MQTT with JSON batch transmission to prevent data loss on the local network.
- Developed the backend with FastAPI + PostgreSQL (SQLAlchemy ORM), featuring JWT/bcrypt authentication and role-based access control (Admin, Doctor, Patient).
- Designed a React frontend with real-time visualization of oscillations and patient grip force.
- Containerized the system's 3 layers (Nginx, API, database) with Docker Compose for reproducible, single-command deployment.

Systems and Communications I · IoT Monitoring and Agro-Climate Analytics

Individual project · Universidad ICESI

- Developed an end-to-end IoT monitoring system for predictive analysis and anomaly detection in coffee and oil palm crops.
- Simulated climate telemetry (temperature, humidity, precipitation, pH) over MQTT and built a Node-RED processing gateway with agronomic range validation, persisting data in InfluxDB.
- Implemented a hybrid predictive model combining incremental learning (PassiveAggressiveRegressor) with periodic retraining (HistGradientBoostingRegressor), weighting predictions by historical MAE.
- Built a real-time visualizer (Matplotlib) with threshold-based anomaly detection.

EDUCATION

B.Sc. Systems Engineering · Universidad ICESI *Jan 2024 – Oct 2029 (exp.)*

Currently in 6th semester. Focus on software development, backend architectures, full-stack development and IoT solutions, with hands-on practice in Java, Spring Boot, Python, SQL and modern frontend technologies.

B.Sc. Telematics Engineering · Universidad ICESI *Jan 2024 – Feb 2029 (exp.)*

Currently in 6th semester. Focus on networking architecture, IoT systems and telecommunications: network design and configuration with Cisco Packet Tracer (IPv4/IPv6, routing, VLANs), MQTT telemetry, Docker containers and MATLAB for modeling and analysis.

Automation Bootcamp · Comfandi Automation Seedbed & Universidad ICESI *July 2026*

Automation Anywhere, cybersecurity, Lean principles and information security.

Programming Fundamentals · Universidad Tecnológica de Pereira *2021*

Technical High School Diploma in Commerce · I.E. Técnica de Comercio Simón Rodríguez *2021*

SENA Certifications in research and laboratory work *2018 – 2020*

CERTIFICATIONS

HackerRank: SQL (Intermediate), SQL (Basic), Java (Basic) and Python (Basic) **Oct 2026**

MathWorks: Telecommunications Onramp **Sep 2026**

ACTIVITIES & INVOLVEMENT

- IEEE member, ICESI Student Branch, since 2024. Represented the branch at the 2026 IEEE National Student Branch Meeting (RNR) in Rionegro, where the branch received multiple awards.
- Active participant in the ICESI Astrobiology Research Group since 2025.
- Member of the ComSoc ICESI student chapter since 2025.
- LINXCO Hackathon (Oct 2026): built Camino Legal in a team of 2.
- GameHub Game Jam by IEEE CIS ICESI (Oct 2026, 24 hours): match backend for the game Beat Potato in Unity and C#, in a team of 5.

SOFT SKILLS

Teamwork, technical leadership, interdisciplinary collaboration, proactivity, independent technical problem-solving, attention to detail and results-oriented mindset.

PERSONAL REFERENCES

- Robeiro Giraldo · Manager, Calipieles · 3206126523
- Abel Guzmán · Coordinator, IETC Simón Rodríguez · 3113672539